

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277616

Kind Code

A1

Publication Date

September 04, 2025

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BLOCK ICE SHAVER WITH TENSION ARM AND SHAPING DEVICE

Abstract

The disclosure extends to devices, systems and methods for producing a shaved ice product using shaved ice confection machines and processes. The disclosure also extends to a tension arm and a shaping device used in conjunction with shaved ice confection machines and processes to produce more consistent shaved ice output with a potential increase in machine life.

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Appl. No.: 19/083112

Filed: March 18, 2025

Related U.S. Application Data

parent US continuation 18053678 20221108 parent-grant-document US 12253291 child US 19083112

parent US continuation 14668835 20150325 parent-grant-document US 10378807 child US 16459447

parent US continuation 13414170 20120307 parent-grant-document US 8939389 child US 14605767

parent US continuation-in-part 16459447 20190701 parent-grant-document US 11493254 child US 18053678

parent US continuation-in-part 14605767 20150126 parent-grant-document US 10215468 child US 14668835

us-provisional-application US 61970349 20140325

Publication Classification

Int. Cl.: F25C5/12 (20060101); F25C5/02 (20060101); F25C5/04 (20060101)

U.S. Cl.:

CPC F25C5/12 (20130101); F25C5/02 (20130101); F25C5/046 (20130101); Y10S241/17 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/053,678, filed Nov. 8, 2022 (now U.S. Pat. No. 12,253,291) which is a continuation U.S. patent application Ser. No. 16/459,447, filed on Jul. 1, 2019 (now U.S. Pat. No. 11,493,254), which is a continuation of U.S. patent application Ser. No. 14/668,835, filed on Mar. 25, 2015 (now U.S. Pat. No. 10,378,807), which claims the benefit of U.S. Provisional Application No. 61/970,349, filed Mar. 25, 2014, and which is a continuation-in-part of U.S. application Ser. No. 14/605,767 filed on Jan. 26, 2015 (now U.S. Pat. No. 10,215,468), entitled “BLOCK ICE SHAVER,” which is a continuation of U.S. application Ser. No. 13/414,170, filed on Mar. 7, 2012 (U.S. Pat. No. 8,939,389) entitled “BLOCK ICE SHAVER,” which are hereby incorporated by reference herein in their entireties, including but not limited to those portions that specifically appear hereinafter, the incorporation by reference being made with the following exception: In the event that any portion of the above-referenced applications are inconsistent with this application, this application supersedes said portion of said above-referenced applications.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT [0002] Not Applicable.

BACKGROUND

[0003] This disclosure relates generally to machines for preparing shaved ice confectioneries, and more particularly, but not necessarily entirely, to a block ice shaver that is particularly adapted for producing a light, fluffy, finely textured shaved ice, or powdered snow-like flavored food products.

[0004] A variety of machines have been developed, described and are widely known for creating or processing cold deserts and confectioneries by processing ice into more appealing eatable forms, such as snow cones and shaved ice. Such devices produce either ice granules (snow cones) or light, fluffy, finely textured shaved ice for subsequent flavoring using syrups. For consistency, a block of ice can produce more appealing ice shavings than cubed ice, for example.

[0005] Despite the advantages of shaved ice machines that are available in the marketplace, improvements are still being sought. Machines in the marketplace may have limitations such as, cumbersome ice block change procedures, frozen and impacted blades, blades that do not produce light, fluffy, finely textured shaved ice, and may produce inconsistent shaved ice textures because the feeding of an ice block into a blade is inconsistent or because the machine cannot adapt to the changing consistency of the ice as the block of ice begins to warm and melt. Such machines that may use large blocks of ice tend to operate at a relatively slow rate, require significant maintenance, and are incompatible with either home use or large volume use.

[0006] Further, such machines may not be able to accommodate the changes in the texture of the block of ice as the ice begins to warm. As ice begins to warm and changes from a very cold state to a warmer state, the texture and consistency of the ice to be shaved by a machine begins to change. As the ice warms, the texture of the ice begins to degrade and the quality of the shaved ice decreases making it more difficult for the machine to shave the warm ice, or at least the ability to

provide a high quality shaved ice product is decreased because of the interaction between the blade of the machine and the ice block.

[0007] Machines in the marketplace may thus be characterized by several disadvantages that may be addressed by the disclosure. The disclosure minimizes, and in some aspects eliminates, the above-mentioned issues, and other problems, by utilizing the methods and structural features described herein.

[0008] The features and advantages of the disclosure will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by the practice of the disclosure without undue experimentation and are readily apparent to those of ordinary skill in the art upon review of the following drawings, detailed description, claims and abstract of this disclosure. The features and advantages of the disclosure may be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Non-limiting and non-exhaustive implementations of the disclosure are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified. Advantages of the disclosure will become better understood with regard to the following description and accompanying drawings where:

[0010] FIG. 1 illustrates a side, perspective view of an embodiment of an ice shaving machine made in accordance with the teachings and principles of the disclosure;

[0011] FIGS. 2A and 2B illustrate the loading of an ice block into an embodiment of an ice shaving machine made in accordance with the teachings and principles of the disclosure;

[0012] FIG. 3 illustrates a schematic view of several components of an embodiment of an ice shaving machine made in accordance with the teachings and principles of the disclosure;

[0013] FIG. 4 illustrates an embodiment of a blade holder in accordance with the teachings and principles of the disclosure;

[0014] FIG. 5 illustrates a cut away view along section AA-AA in FIG. 4 of an embodiment of a blade holder in accordance with the teachings and principles of the disclosure;

[0015] FIG. 6 illustrates a schematic, top view of an embodiment of a blade in accordance with the teachings and principles of the disclosure;

[0016] FIG. 7 illustrates an embodiment of a method and associated structures that allow the rotation of a blade relative to a blade holder in accordance with the teachings and principles of the disclosure;

[0017] FIG. 8 illustrates the concept of tensioning of an embodiment of an ice shaving blade within a blade assembly in accordance with the teachings and principles of the disclosure;

[0018] FIG. 9 illustrates an embodiment of a tensioned blade attached to a blade holder and forming a blade assembly in accordance with the teachings and principles of the disclosure;

[0019] FIG. 10 illustrates the feature of a flexible blade in accordance with the teachings and principles of the disclosure;

[0020] FIG. 11 illustrates an embodiment of a blade holder having enhanced features in accordance with the teachings and principles of the disclosure;

[0021] FIG. 12 illustrates a schematic view of an embodiment of an ice shaving machine having a single drive shaft for actuating the ice block feeder in accordance with the teachings and principles of the disclosure;

[0022] FIG. 13 illustrates an embodiment of a blade in accordance with the teachings and principles of the disclosure;

[0023] FIG. 14 is a detailed view of the embodiment of a blade illustrated in FIG. 13 illustrating

the cutting surfaces and slots in accordance with the teachings and principles of the disclosure; [0024] FIGS. **15A** and **15B** illustrate a side view and a side cross-sectional view taken along M-M of FIG. **15A** of an embodiment of a blade in accordance with the teachings and principles of the disclosure;

[0025] FIGS. **16A**, **16B** and **16C** illustrate top views of various blade embodiments in accordance with the teachings and principles of the disclosure;

[0026] FIG. **17** illustrates an embodiment of a safety clearing tool for use with an ice shaving system disclosed herein and in accordance with the teachings and principles of the disclosure;

[0027] FIG. **18** illustrates a side view of an embodiment of an ice shaving machine having a housing partially lifted up with an ice block guide or a tension arm thereon and made in accordance with the teachings and principles of the disclosure;

[0028] FIG. **19** illustrates a side view of an embodiment of a housing with an ice block guide or a tension arm made in accordance with the teachings and principles of the disclosure;

[0029] FIG. **20** illustrates a schematic view of an embodiment of a housing with an ice block guide or a tension arm that resists rotational forces of a block of ice that is being shaved by a rotating blade and made in accordance with the teachings and principles of the disclosure;

[0030] FIG. **21** illustrates a schematic view of an embodiment of a housing of an ice shaving machine with an ice block guide or a tension arm made in accordance with the teachings and principles of the disclosure;

[0031] FIG. **22** illustrates an embodiment of a shaping device made in accordance with the teachings and principles of the disclosure;

[0032] FIG. **23** illustrates an embodiment of a shaping device attached to a spout assembly of an ice shaving machine in accordance with the teachings and principles of the disclosure;

[0033] FIG. **24** illustrates a bottom view of an implementation of a shaping device for producing a shaved ice or snow cone product in accordance with the teachings and principles of the disclosure;

[0034] FIG. **25** illustrates a side view of an implementation of a shaping device for producing a shaved ice or snow cone product in accordance with the teachings and principles of the disclosure;

[0035] FIG. **26** illustrates a top view of an implementation of a system comprising a shaping device for producing a shaved ice or snow cone product in accordance with the teachings and principles of the disclosure; and

[0036] FIG. **27** illustrates a side view of an implementation of a system comprising a shaping device for producing a shaved ice or snow cone product in accordance with the teachings and principles of the disclosure.

DETAILED DESCRIPTION

[0037] The disclosure extends to machines for preparing shaved ice or snow cone confectioneries. The disclosure also extends to an ice block guide or a tension arm used to provide a force on a block of ice to assist in maintaining the stability of the ice block while rotational forces acting upon the ice block as it is being shaved by a rotating blade. The disclosure also extends to a shaved ice or snow cone shaping device and an attachment mechanism for use with machines that prepare shaved ice or snow cone confectionaries or products.

[0038] Before the apparatus, system and methods for shaving ice, such as blocks of ice, are disclosed and described, it is to be understood that this disclosure is not limited to the particular configurations, process steps, and materials disclosed herein as such configurations, process steps, and materials may vary somewhat. It is also to be understood that the terminology employed herein is used for the purpose of describing particular embodiments only and is not intended to be limiting since the scope of the disclosure will be limited only by the appended claims and equivalents thereof.

[0039] In the following description of the disclosure, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration specific implementations in which the disclosure may be practiced. It is understood that other

implementations may be utilized and structural changes may be made without departing from the scope of the disclosure.

[0040] In describing and claiming the disclosure, the following terminology will be used in accordance with the definitions set out below.

[0041] It must be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise.

[0042] As used herein, the terms “comprising,” “including,” “containing,” “characterized by,” and grammatical equivalents thereof are inclusive or open-ended terms that do not exclude additional, unrecited elements or method steps.

[0043] As used herein, the term “motor” refers to a power source that imparts torque; or any rotating pneumatic or air motor, which does mechanical work by expanding compressed air; or any electromagnetic device used to convert electrical energy into mechanical energy; whether or not the power source, motor or electromagnetic device is housed within or as part of another device.

[0044] As used herein, the term “proximal” shall refer broadly to the concept of a nearest portion.

[0045] As used herein, the term “distal” shall generally refer to the opposite of proximal, and thus to the concept of a further portion, or a furthest portion, depending upon the context.

[0046] The disclosure relates to an ice shaving machine for receiving a block of ice and thinly slicing or shaving the block of ice to produce a powdery snow like textured confection. It should be noted that for the purposes of this disclosure an ice block is defined as a generally homogenous solid body or mass of ice having a volume that is greater than or equal to eight cubic inches, or that is larger than a typical household ice cube that may be placed inside of a user's cup to cool a drink. The ice block may be placed in a feeder of the ice shaving machine, which permits the block to come into contact with a blade of the ice shaving machine. The blade shaves off paper thin slices of ice to create a shaved ice product having a fine snow like texture. A collector collects the paper thin slices of ice and moves it to a spout of the ice shaving machine, under which a container is placed to receive the shaved, slices of ice. Once the container is sufficiently full, flavorings may be added, either through an integrated flavor dispensing system or from an individual flavor container or from a stand-alone flavor station having a plurality of containers and a plurality of dispensers for dispensing the flavorings. This disclosure comprises various embodiments and alternative blade designs, which are described in detail in the following detailed discussion.

[0047] Referring now to the figures, FIG. 1 illustrates a side, perspective view of an embodiment of an ice shaving machine **100** made in accordance with the teachings and principles of the disclosure. The machine **100** comprises a cabinet **114**, which may be made of various materials, including stainless steel, a light weight fiberglass material, or a plastic material without departing from the scope or concept of the disclosure, for durability and ease of cleaning. It will be appreciated that various embodiments of the machine **100** may comprise one or more of the following features: a blade assembly **310**, including a blade holder **301** and a blade **101**; a safety cover **102a** and a blade cover assembly **102b**, including a spout assembly **105**, a lid **111** and an ice block guide **113**; an ice block feeder **103**, including at least one drive shaft **116**; a main drive motor **305** for rotating the blade assembly **310** (illustrated best in FIG. 3); and at least one motor **316** for actuating the at least one drive shaft **116** (illustrated best in FIG. 3). The machine **100** may also comprise a flavor dispenser **104**.

[0048] Continuing to refer to FIG. 1, the safety cover **102a** may be attached to the top of the cabinet **114** for receiving a block of ice **107** and covering the working parts of the machine **100**, for example the blade assembly **310**. The safety cover **102a** may be manufactured from a generally clear “lexan” plastic material, although other sufficiently hard materials to protect a user and act as a safety cover may also suffice and are meant to fall within the scope of the disclosure. The safety cover **102a** may have a base portion **109** for covering the blade assembly **310**. The base portion **109** may be an outer shell fitting over and around the blade assembly **310**, including the blade **101** and blade holder **301**.

[0049] It will be appreciated that the spout assembly **115** may be a separate component from the cover **102a**, and may be part of the blade cover assembly **102b**. The spout assembly **115** may comprise a spout **105** that may be domed shaped or otherwise shaped for providing a domed round top or other shape of shaved ice for the produced shaved ice product. The spout assembly **115** may also comprise an outer shell or base portion **109** fitting over and around the blade **101** and the blade holder **301**. The spout **105** and the outer shell or base portion **109** may be in mechanical communication with each other, such that shaved ice may move from the interior of the outer shell or base portion **109** through a conduit **105a** to the spout or opening **105** for dispensing the shaved ice into a container (illustrated best in FIG. 1). It will be appreciated that the spout **105** may be shaped other than as a dome and may be part of, or separate from, the safety cover **102a** and even the spout assembly **115**. For example, the spout **105** may be in mechanical communication with the base portion **109** of the spout assembly **115**, or the spout **105** may be attached directly to the cabinet **114** or other structural member, such that the spout **105** is not in mechanical communication with the spout assembly **115**, but acts for creating a shaped shaved ice product, without departing from the scope of the disclosure.

[0050] Referring now to FIGS. 2A and 2B, the spout assembly **115** may also comprise a flap **117** that may be a flexible material, such as a plastic or other material that is durable, but pliable, to help and assist a user in making a shaved ice confection. The flap **117** may be used as a part of the spout assembly **115** or may be a separate and distinct component. The flap **117** may be used as a guide to assist in the dispensing of the shaved ice, such that less shaved ice is lost as the shaved ice is dispensed from the spout **105** to the container. The flap **117** may at least partially wrap around the container and may be used, or operated, as a chute so that shaved ice is directed into the container as the shaved ice is dispensed from the spout **105** or other opening.

[0051] Mounted to the safety cover **102a** is a lid **111** that may be movably fastened to the safety cover **102a**. In an embodiment, an electrical or mechanical switch **112** is provided in the lid **111** safety cover **102a** connection to shut off the operation of the machine **100** when the lid **111** is raised. The lid **111** may further comprise an ice block guide **113** that adds a biasing force laterally to the ice block so that the ice block will be more consistently fed into the blade **101**. The ice block guide **113** may be a simple compliant type structure or may have a plurality of parts that produce a biasing force. The ice block guide **113**, also referred to herein as a tensioning arm **113**, may provide a force to assist in maintaining the stability of the ice block and resist rotational forces acting on the ice block as it is being shaved by a rotating blade **101**.

[0052] Referring to FIGS. 2A-3, the ice block feeder **103** may comprise, or may be mounted or otherwise attached to, a drive shaft **116**. The ice block feeder **103** may operate to contact the block of ice **107** and place a force on said block of ice **107**. The drive shaft **116** operates to actuate the ice block feeder **103** into contact with the ice block **107**, thereby bringing and forcing the ice block **107** into contact with the blade **101**. The ice block feeder **103** may comprise or may be mounted or otherwise attached to a single drive shaft **116** or, alternatively, to two or more drive shafts **116**. When two or more drive shafts **116** are utilized, the drive shafts **116** work in unison to force, move and push the ice block **107** into contact with the blade **101** to produce shavings of ice. The ice block feeder **103** may continuously move the ice block **107** into the blade **101**, freeing flakes or shavings of ice, which are received by the spout **105** of the blade cover assembly **102b**.

[0053] It will be appreciated that the blade **101** may be generally planer, generally circular, substantially flat and slotted with a plurality of slots or openings **151** (shown best in FIGS. 6, 13-16C). However, it will be appreciated that the blade **101** may be another geometric shape, besides circular, without departing from the scope of the disclosure. Each slot or opening **151** may have an adjacent edge **150** that may be slightly bent upwards toward the direction of the incoming ice block. This blade **101** may be made of high quality stainless steel material to maximize the blade life and reduce corrosion. Other non-corrosive high strength materials may also be used in place of stainless steel without departing from the scope of the disclosure. The blade **101** may be affixed to

a blade holder **301** to form a blade assembly **310**, such that the assembly can be rotated to provide more consistent ice shavings. While in use, the ice shavings fall through the blade **101** by way of the slots **151** and into a collection area where the shavings are moved toward the spout **105** and are then available to be dispensed as desired into a container. Once a sufficient quantity of ice shavings are collected in a container, flavoring may be added as desired.

[0054] Referring back to FIG. 1, a flavor dispenser **104** may be provided as part of the shaving device. In an embodiment of the disclosure **100**, flavor selection switches may be provided on the front of the cabinet **114**. The flavors may be provided by a plurality of inverted bottles or other containers (not shown), which may be fitted to receptacles on the top of the cabinet **114**. An automated water switch may also be provided to flush or clean the flavoring from the dispenser system. The water can also be used to dilute the concentrated flavoring. A drain pan may be provided with a drain, which may have a drain hose for handling spills.

[0055] Referring back to FIGS. 2A and 2B, an embodiment of a block ice shaving machine **100** is illustrated as it is being loaded and used. During use, a user may slide the lid **111** in an upward direction to an open position. As can be seen in FIG. 2A, this movement is illustrated by the upward pointing arrow. In the open position, access to the ice block feeder **103** is provided and the ice block **107** can be loaded into position over the blade **101**. It will be appreciated that the structural features of the safety cover **102a** permit a user to gain access to the ice block **107**, even after the ice block **107** has been loaded into the machine **100**. The ability to access the block of ice **107** after being loaded into the machine **100** may be advantageous for several reasons, including the ability to clear blocked shaved ice from the blade **101** or the spout assembly **115** or removing unused portions of the block of ice and potentially others. Some machines that are available in the market do not allow or otherwise permit access to the ice once it is loaded into the machine. The result of those machines is that time and effort may be wasted while the machine either works to clear itself of frozen or otherwise stuck shaved ice, or simply shaves the excess ice that is loaded in the machine as waste product.

[0056] The ice block feeder **103** may be raised automatically or manually to allow the insertion of an ice block **107** into an opening or space between the ice block feeder **103** and the blade **101**. An electronic switch may be provided to control the position and travel of the ice block feeder **103**. The electronic switch, whether a toggle switch or a rocker switch, may control at least one electric motor **316** (illustrated best in FIG. 3) in order to raise and lower the ice block feeder **103**. A voltage regulator/transformer may receive AC power via a standard three prong power cord and produces DC electric power for the motor **316**. Between the voltage regulator/transformer and the motor **316** may be a speed controller, which includes a motor speed selection dial (not shown). The voltage regulator/transformer may be connected via an electric connection to the speed controller, which in turn, is connected to the motor **316**. The motor may be a DC motor capable of turning the shaft at between 400 and 500 rpm. Alternatively, it is contemplated that an AC motor capable of 1100 rpm may be substituted. Other motor configurations can be substituted without departing from the spirit or scope of the disclosure. The motor **316** may be connected to one or more drive shafts **116** to drive the ice block **107** into the blade **101** at a predetermined rate and at a predetermined pressure or force.

[0057] One embodiment of the ice block feeder **103** is one that drives the ice block **107** at a range within about 180 lbs. to about 220 lbs. of pressure. In another embodiment, the ice block feeder **103** may apply more or less pressure than previously stated in the range, and may further use gravity and the mass of the ice block **107** itself to feed the ice block **107** into the blade **101**.

[0058] As can be seen in FIG. 2B, the lid **111** may be rotated to lock into an open position for convenience. Other mechanisms may be employed to lock the lid **111** in an open position and are contemplated to be within the scope of the disclosure.

[0059] FIG. 3 illustrates a schematic view of an ice block shaved ice machine **100**. In an embodiment, the block ice shaving machine **100** may comprise the main drive motor **305** that may

be mechanically coupled or connected to the blade assembly **310**, or directly to the blade **101**, and configured for shaving ice from an ice block. The mechanical coupling may be accomplished through additional structures, such as gearing and pulleys or by blade holders **301**. A control mechanism for activating and deactivating the mechanical drive mechanism, such as the main drive motor **305**, may be used to operate the machine **100**. For example, the control mechanism may comprise a switch device to control the main drive motor **305**, which may be actuated by actuating the switch device. The switch device may be a toggle switch or a rocker switch, which may advantageously provide a safety mechanism such that there is substantially no chance of accidental bumping of the switch into an activated position. The main drive motor **305** may also be actuated using a foot pedal switch.

[0060] In an embodiment, the blade **101** may be attached to the blade holder **301**, and it is the blade holder **301** that may be directly attached to the drive shaft **307** of the main drive motor **305**. A blade holder **301** and blade **101** may be assembled together using complementary structures to form the blade assembly **310**. It will be appreciated that the blade **101** may be attached or otherwise connected to the blade holder **301** using a variety of different mechanical structures. For example, threaded inserts and fasteners, screw fasteners, key and keyhole fasteners, post or prong style snap fasteners, mechanical pins, clips or hooks, anchors, bolts, clamps, locknuts, rivets, screws and washer assemblies, or other fasteners known or that may become known in the future may be used to attach or connect the blade **101** to the blade holder **301** without departing from the scope of the disclosure.

[0061] In an embodiment, the ice block feeder **103** of the shaved ice machine **100** may comprise, or may be connected to, one or more drive shafts **116**. The drive shafts **116** may operate by way of a screw drive rod **317** that is driven by an ice block feeder electric motor **316**. The drive shafts **116** may be connected one to another with a push plate **303** that actually makes contact with the block of ice **107**. Electronic controls may be employed to control any block feeder electric motors **316** as well as the main drive motor **305**. In an embodiment, the main drive motor **305** may be mechanically coupled to the blade assembly **310** through a transmission or gear box (not shown) having predetermined operational inputs and outputs. As can be seen in FIG. **12**, it is within the scope of the disclosure to contemplate an ice block shaving machine **1200** having less than two drive shafts **116**, and it will be appreciated that the disclosure also contemplates a machine that has more than two drive shafts **116**. The ice block shaving machine **1200** may have a single drive shaft **116** and may have additional structures to allow the successful use of a single drive shaft **116**, such as a single motor **316**.

[0062] Referring now to FIG. **4**, there is illustrated a top down view of the blade holder **301** of the blade assembly **310**. In the embodiment shown, it can be seen that the blade holder **301** may comprise a plurality of arms **404**. The drawings of this disclosure illustrate, for exemplary purposes only, embodiments having three or four arms, but it will be appreciated that the plurality of arms **404** may include two, three, four or more arms without departing from the scope of the disclosure.

[0063] In an embodiment, and as illustrated in FIG. **4**, the blade holder **301** may comprise a plurality of arms **404**, for example three arms **404**. The plurality of arms **404** may each radiate out from a central hub **406**. At some distance from the center hub **406**, a fastening or attachment structure **410** may be disposed on each of the plurality of arms **404**. The fastening or attachment structure **410** may correspond with an attachment point (such as **615** in FIG. **6** or **706** in FIG. **7**) of the blade **101**. The fastening or attachment structure **410** may be used to fasten the blade **101** to the blade holder **301**.

[0064] It will be appreciated that a radial measurement from a center of rotation of the blade holder **301**, which center of rotation may be the center of the hub **406**, to the attachment structure **410** of each of the plurality of arms **404** is less than a radial measurement from a center of rotation of the blade **101** to the attachment points (such as **615** of FIG. **6**) of the blade **101**. The result may be that the blade **101** is in a tensioned state as discussed more fully below with respect to FIGS. **8** and **9**.

[0065] The center hub **406** may comprise a blade support that mechanically communicates and interacts with the blade to control the amount of flex of the blade as a load is placed on the blade. The result may be that the blade is loaded evenly thereby reducing wear and tear of the blade, which may extend the life of the blade.

[0066] In an embodiment, the blade **101** may be attached to the blade holder **301**, and it is the blade holder **301** that is directly attached to the drive shaft **307** of the main drive motor **305**. It should be noted that the blade holder **301** may be directly attached to a drive shaft of a motor with any manner of known means, such as complementary threaded portions, set screws or any other means known or yet to be discovered. With regard to the number of arms **404** of the blade holder **301**, it may be advantageous to have enough arms **404** to control forces on the blade **101** and resist undesired rotation. For example, three arms **404**, as illustrated in FIG. 4, would provide the desired stability in the typical “x” and “y” planes, such that the machine **100** will work relative to a single axis of rotation. It will be appreciated that the single axis of rotation may be a vertical axis or a horizontal axis of rotation. It will be appreciated that the single axis of rotation may be coaxial with a shaft, such as **307**, of a drive motor, such as **305**, in a simple embodiment. Alternatively, the single axis of rotation may not be coaxial or even parallel to a shaft, such as **307**, of a drive motor, such as **305**, if gearing is to be used in the machine.

[0067] Referring now to FIG. 5, the figure illustrates a cut-a-way view of a blade holder **301** defined by the cut lines AA-AA as shown in FIG. 4. In the cut-a-way view, greater detail regarding the center hub **406** is shown and which comprises a driver interface **513** having threads **514** that are schematically shown for this embodiment. A drive shaft (not shown) of an appropriate electric motor (not shown) may be inserted into the driver interface **513** and may be affixed by threading the threads **514** of the blade holder **301** onto corresponding threads on said drive shaft. The threads may be reverse threaded or traditionally threaded such that during use the blade holder **301** does not separate from the drive shaft. The blade holder may also be held with a set screw or key slot structure.

[0068] As can further be seen in FIG. 5, the plurality of arms **404** may comprise additional structures and structuring in order to provide desired functionality. For example, as illustrated, a pass-through portion **525** may be provided that allows ice shavings to pass over the top of the arm **404** yet under the blade **101**, such that the ice shavings can freely move and be forced into a collection area and out of the spout **105**. The pass through portion **525** may be defined by the attachment structure **410** on one end and the hub **406** on the other end as illustrated in FIG. 5. The pass through portion **525** allows the ice shavings to easily pass over the top of the arm **404** because of the height of the material forming the arm **404**. As illustrated in the embodiment, the arm **404** may be formed by a sloping or tapered portion **404a** adjacent the hub **406**, which tapers in a proximal-to-distal direction with respect to the hub **406**. The tapered portion **404a** is immediately followed by a substantially planar portion **404b**. It will be appreciated that there may be a pass-through portion **525** for each of the plurality of arms **404**.

[0069] Accordingly, while ice is being shaved from a block of ice, the shavings will be allowed to fall through the slots or openings **151** in the blade **101** and into a collection area that is away from the blade **101** at all points under the blade **101**. It will be appreciated that the greater the surface area present on the arm, such as **404** or **1104**, the more potential there is for ice build-up on the arm. Thus, it is contemplated by the disclosure to have an arm **404** that may have a small surface area, thereby creating a larger pass through portion **525**, or an arm **404** that may have a large surface area as illustrated in FIG. 5 thereby creating a smaller pass through area **525**. In an embodiment, the arm may be relatively petite in comparison to the hub. The petiteness of the arm may allow any ice that has adhered to the material that comprises the arm to easily break off as the arm rotates, thereby continuing to permit ice shavings to pass over the arm and through the pass through portion **525**.

[0070] A potential advantage of the pass through portion **525** is the ability for the ice shavings to

move freely without substantial sticking, adhering, freezing to the arm of the blade holder or otherwise clogging up the machine. Allowing free movement of the ice shavings prevents clogging and freezing problems. The surface of the arms **404** may be textured or coated for decreasing icing problems. The arms **404** may further comprise a paddle portion **528** that, together with the arm structure, may help to control the movement of the ice shavings within the machine. The arms **404** may further comprise a fastening structure **410**, which may include threaded openings, that may be configured to accommodate fasteners therein to attach the blade **101** to the blade holder **301** to form the assembly **310**.

[0071] Additionally, as can be seen in FIG. **11**, the arms **1104** of blade holder **1110** may be curved as they radiate out from the central hub **1106**. The curvature of the arms **1104** may function and work to reduce the accumulation of ice shavings on the arm **1104**. It will be appreciated that the radius of curvature of one or more of the plurality of arms **1104** may provide for a greater or lesser reduction of the accumulation of ice shavings on that arm **1104**.

[0072] Still further, it will be appreciated that the disclosure contemplates an embodiment wherein the arm may have a very large surface area, such that there is a very little pass through portion or substantially no pass through portion at all. In such an embodiment, substantially the entire arm operates and acts as a paddle to help move or push the shaved ice into the conduit **105a** and out of the spout assembly **115** as the blade holder rotates.

[0073] Referring now to FIG. **6**, there is illustrated a schematic view of a blade **101** consistent with the teachings and principles of the disclosure. The blade **101** may be generally circular in shape and be configured to rotate about an axis that runs geometrically normal to a top surface **170** of the blade **101** and passes through the center point of the circular blade **101**. However, it will be appreciated that the blade **101** may be another geometric shape, besides circular, without departing from the scope of the disclosure. The blade **101**, may be generally planer, substantially flat and slotted with a plurality of slots or openings **151**.

[0074] As illustrated in FIGS. **6**, and **13-15B**, there may be a significant number, perhaps hundreds, of slots or openings **151**. Each slot or opening **151** may have an adjacent cutting edge **150** that may be slightly bent upwards toward the direction of the incoming ice block. In FIGS. **15A** and **15B**, which illustrate a side view and a side cross-sectional view of an embodiment of a blade in accordance with the teachings and principles of the disclosure, the cutting edge **150** is illustrated as pointing downward with respect to the page. However, it will be appreciated that the ice block may be forced into the cutting edge **150** or blade **101** as described herein. Thus, the blade **101** illustrated in FIGS. **15A** and **15B** may be oriented with respect to the ice block, such that the cutting edge **150** or blade **101** is positioned to cut into the ice block.

[0075] The blade **101** may be made of high quality stainless steel to maximize the blade life and reduce corrosion. Other non-corrosive high strength materials may also be used in place of stainless steel without departing from the scope of the disclosure. The blade **101** may be affixed to the blade holder **301** to form the blade assembly **310** such that the assembly can be rotated by a drive motor or other mechanical drive mechanism to provide more consistent ice shavings. For example, a hand crank or other mechanical drive mechanism may be used to rotate or drive the blade assembly without departing from the scope of the disclosure. While in use, whether through a drive motor or other mechanical drive mechanism, the ice shavings fall through the blade **101** by way of the slots **151** and into a collection area where the shavings are moved through a conduit toward the spout **105** and are then available to be dispensed as desired into a container. Once a sufficient quantity of ice shavings are collected in a container, flavoring may be added as desired.

[0076] The blade **101** may comprise a cutting portion **610**, which may include the slots or openings **151** and the cutting edges **150**, and a fastening portion **611**. The fastening portion **611** may comprise radially placed attachment points **615**. One or more of the attachment points **615** may be located around the perimeter of the blade **101**. The attachment points **615** may be any structure that facilitates the attachment of the blade **101** to the blade holder **301** and may correspond with the

fastening or attachment structure **410**. In an embodiment, the attachment points **615** may be openings or holes that fasteners pass through and into the fastening structure **410** to secure the blade **101** to the blade holder **301**. However, it will be appreciated that other fastening mechanisms may also be used without departing from the scope of the disclosure. For example, threaded inserts and fasteners, screw fasteners, key and keyhole fasteners, post or prong style snap fasteners, mechanical pins, clips or hooks, anchors, bolts, clamps, locknuts, rivets, screws and washer assemblies, or other fasteners known or that may become known in the future may be used to attach or connect the blade **101** to the blade holder **301** without departing from the scope of the disclosure.

[0077] As can be seen in FIGS. **13-15B**, an embodiment of the blade **101** can be seen that illustrates the adjacent placement of the raised cutting surfaces **150** relative to the slot or openings **151**. FIG. **14** shows a detailed view of a portion of an embodiment of a blade **101** of FIG. **13**. It should be noted that the ice shaving blade **101** may be made of many different and various configurations as far as the cutting surfaces **150** and slot **151** patterns are concerned. For example, FIGS. **16A**, **16B**, and **16C** illustrate various embodiments of the blade **101** and examples of patterns created by the cutting surfaces **150** and slots or openings **151**, which may be utilized by the disclosure.

[0078] It will be appreciated that a ratio of the surface area of the material from which the blade **101** is manufactured to the slot or opening area may be tailored or predetermined for desired shaved ice output and machine functionality. For example, the ratio may be about 25% material to about 75% opening, or the ratio may be inverted, such that there is about 75% material to about 25% opening. It will be appreciated that any ratio falling within the ranges given above are meant to fall within the scope of the disclosure, for example, the ratio may be 40% material to about 60% openings or about 60% material to about 40% openings.

[0079] An embodiment may have a blade **101** that corresponds to a blade holder **301** in which the blade **101** has more attachment points **615** than the blade holder **301** has arms **404**. An embodiment may comprise a blade holder **301** having the same number of arms **404** as the blade **101** has attachment points **615**. In certain embodiments, the blade **101** may be configured such that it flexes during use. Accordingly, in such embodiments the portions of the blade **101** that are adjacent to the arms **404** of the blade holder **301** may be more rigid than portions of the blade **101** that are further away from the support arms **404** of the blade holder **301**. Consequently, the latter, less supported portions of the blade **101** may flex away from the block of ice during use and thus experience less wear and tear. Accordingly, it may be advantageous to rotate the blade **101** relative to the blade holder **301** (and affix the blade **101** to the blade holder **301** in an alternate position) thereby distributing the wear and tear across the whole blade surface, thereby more evenly wearing the cutting surfaces **150** of the blade **101** resulting in longer blade life.

[0080] Referring now to FIG. **7**, there is illustrated a method and associated structures that allow the rotation of a blade relative to a blade holder. The blade **701** may be shifted or rotated with respect to the blade holder **710** in order to prolong the life of the blade **701**. As can be seen in the figure, the blade holder **710** may comprise a plurality of arms **704**, for example, three arms **704**, and is shown being shifted or rotated in dashed lines relative to the blade **701** in order to prolong the usable life of the blade cutting surfaces. More specifically, the arms **704a** may be positioned in position "A" in the figure, which position represents a first assembled position of the blade assembly **700**. The arms **704b** may be positioned in position "B" in the figure, which represents a second assembled position of the blade assembly **700**. As a user determines that the blade **701** is performing in an unsatisfactory manner, the blade **701** can be rotated relative to the blade holder **710** to improve function. The blade **701** may be rotated incrementally one attachment point **706** at a time or may be rotated by a plurality of attachment points **706** in order to utilize a different part of the blade cutting surfaces that have not been used as much as other portions of the blade to extend the life and usefulness of the blade as long as possible.

[0081] FIGS. 8 and 9 illustrate the feature of a tensioned blade **801** in a blade assembly **800**. In order to reduce the accumulation of unwanted ice, or ice build-up, on the blade **801**, the blade **801** may be tensioned during the assembly of the blade assembly **800**. As can be seen schematically in FIG. 8, the blade **801a** may be relatively planer prior to assembly while it is in an un-tensioned state. The blade holder **810** may be configured with a distance between the mounting holes **815** on the blade holder arms **804** that is less than the distance between corresponding attachment points **816** of the blade **801**.

[0082] It will be appreciated that a radial measurement or distance D1 (illustrated in FIG. 9 and FIG. 11a) from a center of rotation of the blade holder **301** (depicted as dashed line R-R in FIG. 9), which center of rotation R-R may be the center of the hub **406** (see FIG. 9), to the center of the attachment structure **410** of each of the plurality of arms **404** is less than a radial measurement or distance D2 (illustrated in FIG. 6) from a center of rotation R2 of the blade **101** to the attachment points (such as **615** of FIG. 6) of the blade **101**. The radial measurement D1 may be between a range of about two inches to about eight inches, or may be between about three inches to about seven inches, or may be between about four inches to about six inches, or may be about five inches. It will also be appreciated that distances or lengths of the radial measurements D1 may be modified depending upon the size of the blade **101** and blade cover assembly **102b** to be used with the machine **100**. It will be appreciated that the distance or length of a diameter D3 of the blade holder **1110** may be approximately twice the length or distance of the radial measurement D1, such that the diameter may be between about four inches to about sixteen inches, or may be between about six inches to about fourteen inches, or may be between about eight inches to about twelve inches, or may be about ten inches. Due to the above mechanical properties and physical relationships, and as shown by the blade **801b** in FIG. 8, the blade **801** will be placed in tension as and when it is attached to the blade holder **810**. As illustrated in FIG. 9, the blade **801** is in a tensioned state when it is attached to the blade holder **810**.

[0083] Referring briefly now to FIGS. 5 and 11, the fastening or attachment structure **528** of the blade holder **301** may extend upwardly from the arm **404**. It will be appreciated that an angle may be formed between the arm **404** and the attachment structure **528** at an angle that is substantially normal to the plane in which the arm **404** generally lies, such that the attachment structure **528** may be aligned with the attachment points, such as **706**, of the blade, such as **701**, so that a fastener or other mechanical closure may secure the blade to the blade holder **301** to thereby form the blade assembly **310**. The attachment structure **528** may comprise a height H1. The height H1 may be about 0.500 inch to about 3.0 inches, or may be about 1.0 inch to about 2.0 inches, or may be about 1.50 inches.

[0084] It will be appreciated that the hub **406** may also comprise a height H2, which may have a value that is less than, equal to, or larger than the height H1 of the attachment structure **1110**. Thus, for example, the height H2 may be about 0.500 inch to about 3.0 inches, or may be about 1.0 inch to about 2.0 inches, or may be about 1.50 inches, when the height H2 of the hub **406** is about equal to the height H1 of the attachment structure **528**. The height H2 of the hub **406** may also be about 0.650 inches to about 3.150 inches, or may be about 1.150 inches to about 2.150 inches, or may be about 1.650 inches, when the height H2 is larger than the height H1. A ratio of the height H2 of the hub **1106** to the height H1 of the attachment structure **1110** may be about 0.8 to about 1.5, or about 1.0 to about 1.5 or may be about 1.1 to about 1.3. The difference in height values, or lack thereof, between H1 and H2 may result in a blade that can flex to a predetermined extent until contacting the hub **1106**. For example, when the height H1 is substantially equal to the height H2, the blade may be able to flex to a greater degree than when H2 is larger than H1.

[0085] Referring now to FIG. 10, the figure illustrates the feature of a flexible blade during use. In the figure it can be seen that as a block of ice **1007** is pushed into the spinning or rotating blade **1001**, the blade flexes a distance represented in the figure as distance "F." In the figure, the ice block positioned and labeled at **1007a** (and drawn in a solid line) represents an unloaded condition

and position. As a load is applied to the ice block **1007a** by an ice feeder **103** (illustrated best in FIGS. **1-3**), the blade **1001a** flexes to the blade positioned and labeled at **1001b** (and drawn in dashed lines), which shows a loaded and flexed condition. The loads applied by the ice feeder **103** may range from the weight of the ice block itself to a considerable load of about 180 to about 220 lbs. of pressure or more (for example, 250 lbs. of pressure). Variables that can help determine the load to be applied may be such things as: the blade design, the blade material, the shaved ice rate, the limitation of ice feeder power, environmental conditions, the rotation rate of the blade and other factors.

[0086] The hub may comprise a support structure **1009** that mechanically communicates and interacts with the blade to control the amount of flex of the blade as a load is placed on the blade, such that the blade may be loaded evenly. The flex of the blade may be limited by the blade support structure **1009**, which may be part of the hub or other structure, on the blade holder **1010** to prevent damage to the blade **1001**. It will be appreciated that the support structure **1009** may comprise a substantially flat surface or may comprise a curved surface without departing from the scope of the disclosure. It will also be understood that the release of the load on the blade **1001** may allow the blade to move back into its original, unloaded position. When the load is released, the blade moves and experiences a “spring action,” such that any built-up ice may be removed from the cutting surfaces of the blade **1001** as the blade springs back into its original, unloaded position.

[0087] It will be appreciated that the disclosure contemplates an embodiment in which the blade may be attached to the blade holder without being in tension, such that the blade does not flex. In such an embodiment, the blade may be substantially static with respect to the blade holder and hub.

[0088] Referring now to FIG. **17**, there is illustrated a tool for safely clearing blockages within an ice block shaving machine. A clearing tool **1600** may be included in a system comprising an ice shaving machine having a blade assembly.

[0089] FIGS. **18-21** illustrate various views of an ice shaving machine having a housing partially lifted up with an ice block guide or a tension arm **113** thereon in accordance with the teachings and principles of the disclosure. The ice block guide or tension arm **113** illustrated may provide a force on a block of ice to be shaved by the blade **101** of the ice shaving machine **100**. FIGS. **18-20** illustrate the ice block guide or tension arm **113** with the other features and structures disclosed herein. It will be appreciated that the ice block guide or tension arm **113** may resist rotational forces of a block of ice **107** that is being shaved by the rotating blade **101**.

[0090] It will be appreciated that the tension arm **113** may be sized and shaped in various manners. For example, the ice block guide or tension arm **113** may comprise an edge **121**, wherein the edge **121** is configured to contact the block of ice **107**, thereby placing a compressive force on the block of ice **107**. It will be appreciated that the edge **121** may be oriented parallel to the block of ice during shaving, such that the block of ice **107** travels along the edge **121** while being shaved.

[0091] The ice block guide or tension arm **113** may be removably attached to the safety cover **102a** with fasteners or the like. It will be appreciated that the fasteners may be removable without the use of tools to thereby facilitate easy maintenance and installation of the guide or arm **113**. The ice block guide or tension arm **113** may be substantially rigid and manufactured or made of unitary construction and material. The ice block guide or tension arm **113** may be made from a polymer or other suitable material that can provide a force on the ice block without departing from the spirit or scope of the disclosure. The ice block guide or tension arm **113** may comprise a blade portion that comprises the edge **121** that is shaped to optimally contact the block of ice. The blade portion may further comprise a rounded profile or the blade portion may comprise a sharpened edge.

[0092] Referring now specifically to FIG. **21**, the ice shaving machine may further comprise a tensioning mechanism **2100** for adjusting a force that the tensioning arm **113** exerts on the ice block **107**, wherein the tensioning mechanism **2100** contacts a surface **125** of the tensioning arm **113**. The tensioning mechanism **2100** may comprise a threaded member **2110** and a crank **2120** having a handle, which is configured to drive the threaded member **2110**. It will be appreciated that

the crank **2120** may be hand operated, or the crank **2120** may be driven by an electric motor without departing from the scope of the disclosure. Further, it will be appreciated that the tensioning mechanism **2100** may automatically adjust relative to a size of the block of ice **107**, such that as the block of ice melts faster the tensioning mechanism automatically adjusts by placing added force on the block of ice to thereby maintain the block of ice in position for efficient shaving.

[0093] Referring now to FIGS. **22-27**, there is illustrated an embodiment of a shaping device **2200** and a system for shaping shaved ice products made in accordance with the teachings and principles of the disclosure.

[0094] It will be appreciated that a variety of machines have been developed, described and are widely known for creating or processing cold deserts and confectioneries by processing ice into more appealing eatable forms, such as snow cones and shaved ice and the like. Such devices produce either ice granules (snow cones) or light, fluffy, finely textured shaved ice for subsequent flavoring using syrups.

[0095] Machines in the marketplace may have a spout assembly comprising a channel that leads to a spout for expelling shaved ice or snow cone from the machine to a cup or other container for consumption by a user. Such spout assemblies, or spouts that expel shaved ice or snow, are often made from a hard plastic or other rigid material, such that adaptation to different shapes and sizes of cups is often difficult. For example, different cup styles and types, each having their own benefit to a consumer, are becoming increasingly popular. One such shape is a flower cup, but other open faced cups having large mouths or openings for creating a larger container for efficiently contain a shaved ice product are also becoming popular. To effectively fill these different cup sizes and styles with shaved ice, a different size spout may often be required to increase the efficiency (e.g., speed) of filling a cup with shaved ice or ice granules (snow cones).

[0096] In addition, because shaved ice or snow cone machines have an associated cost for each unit it is often expensive to provide multiple machines at one location. Even if multiple machines are financially feasible, changing from one machine to another tends to cause slowness to the overall shaved ice or snow cone operation, which may be disadvantageous in industries (such as shaved ice) where speed is required to keep a consumer line moving. For example, moving between machines or changing out machines may require large amounts of human capital to operate, thereby reducing the efficiency of a business.

[0097] In some instances an operator may use a sanitary glove to shape a shaved ice or snow cone product. The operator may be required to remove or change the sanitary glove after handling money or otherwise exposing the glove to an unsanitary surface. That glove change or removal process may slow down the efficiency of the business.

[0098] In various industries, for example a shaved ice or snow cone business or any other business in the concessions industry, it is important for the success of that business to move customers through a waiting line as quickly as possible to finalize the sale of a confectionary product to customers. In the example of a shaved ice or snow cone business, the ability to quickly modify the spout to fill different sizes and types of cups on the fly at the job site is imperative to the success of a shaved ice or snow cone concession stand, especially in locations where time is of the essence, for example at a halftime break at a sporting event or other intermission. Otherwise, when moving between machines or modifying the spout of the machine consumes too much time the business will lose out on the opportunity to make a sale. This is because the break or intermission is either over or the customers are tired of waiting in long lines and will seek other concessions.

[0099] FIGS. **22-27** illustrate an apparatus, system and method for attaching a device or an assembly to a spout or other part of the shaved ice or snow cone machine in an efficient, effective and elegant manner that will allow a user to fill various size and shapes of cups or containers.

[0100] The shaping device and system for producing a shaved ice or snow cone product are illustrated and may comprise an ice shaving or snow cone machine, including all of its component

parts, a shaping device **2200**, and a blade for producing a shaved ice or snow cone product.

[0101] The shaping device **2200** may be used in conjunction with an ice shaving or snow cone machine **100** (illustrated best in FIGS. **1-2B**, and **18**) for producing a shaved ice or snow cone product. Due to the various shapes and sizes of containers, particularly those with open mouths or large-mouth openings, that may be used to contain a shaved ice or snow cone product, there are times when providing a shaping device **2200** may be advantageous for efficiently filling such containers not only due to ease of use, but also in time savings for filling such containers. Further, the use of the shaping device **2200** in conjunction with various machines for producing a shaved ice or snow cone product may also eliminate the need for a user to utilize his or her hands (whether gloved or otherwise) to shape the shaved ice or snow cone product, thereby making the product more sanitary. The shaping device **2200** may comprise a body **2210** that may be sized and shaped to produce the shaved ice or snow cone product. It will be appreciated that the shaved ice or snow cone product may correspond in shape with the shape of the body **2210** of the shaping device **2200**.

[0102] The body **2210** of the shaping device **2200** may comprise an inner surface **2212** and an outer surface **2213**. The inner surface **2212** of the body **2210** may comprise a textured surface **2214**. The textured surface **2214** may be used to produce a shaved ice or snow cone product having a corresponding textured shape.

[0103] The shaping device **2200** may also comprise a mechanism **2220** for attaching the shaping device **2200** to the ice shaving or snow cone machine **100**. The mechanism **2220** may comprise a connector **2222** that may be configured and dimensioned for communicating with a portion of the ice shaving or snow cone machine **100**. The shaping device **2200** may also comprise a chute **2230** that may be configured and dimensioned for mechanically communicating with a spout **105** or spout assembly **115** of the ice shaving or snow cone machine **100**. The chute **2230** may be in mechanical communication with the body **2210**, such that as ice is shaved via a blade **2201** of the ice shaving or snow cone machine **100**, the shaved ice or snow cone moves through the spout assembly **115** of the ice shaving or snow cone machine **100** into the chute **2230** and into a user's cup or container for carrying or consumption. The chute **2230** may comprise a wall or a plurality of walls that may be sized and shaped to matingly engage and mechanically communicate with a corresponding channel or passageway of the spout **105** or the spout assembly **115**, such that when shaved ice or snow cone is expelled it may be expelled from the spout **105** or spout assembly **115** through the chute **2230** and into the container or cup. It will be appreciated that the body **2210** of the shaping device **2200** may ultimately be used to form the shape of the shaved ice or snow cone product as noted above.

[0104] Referring now to the various mechanisms **2220** for attaching the shaping device **2200** to the ice shaving or snow cone machine **100**, it will be appreciated that in one implementation, the connector **2222** of the shaping device **2200** may be a magnet. In an implementation, the magnet (connector **2222**) may be in magnetic communication with any metallic portion of the ice shaving or snow cone machine **100**. In such an implementation, the connector **2222** does not necessarily have to be in mechanical communication with the spout **105** or spout assembly **115**. Instead, a cup or container may be placed beneath the spout **105** or spout assembly **115** and subsequently matched with the shaping device **2200**. However, it will be understood that it may be advantageous to locate the shaping device **2200** in close proximity to the spout **105** or spout assembly **115**.

[0105] Referring specifically to FIGS. **26-27**, the ice shaving or snow cone machine **100** may comprise an area wherein ice may enter into to be introduced to the blade **2201**. Once the ice is shaved by the blade **2201**, the ice may travel through a channel **2216** before being expelled from the spout **105**. The spout assembly **115** may comprise a corresponding connector mechanism **2240**. In one implementation, the connector **2222** of the shaping device **2200** may be a first magnet **2223** that may be in magnetic communication with the corresponding connector mechanism **2240** of the spout assembly **115**. In such an implementation, the connector mechanism **2240** may be a second magnet **2243** that may be connected to the ice shaving or snow cone machine **100**. In an

implementation, the first magnet **2223** may be attached to a portion of the chute **2230** and the second magnet **2243** may be attached to the spout **105** of the ice shaving or snow cone machine **100**, such that the first magnet **2223** and the second magnet **2243** together hold the shaping device **2200** in location with respect to the spout **105** and spout assembly **115**.

[0106] It will be appreciated that in an implementation the connector **2222** of the shaping device **2200** may be mechanically attachable to a portion of the ice shaving or snow cone machine **100**. For example, the mechanical attachment of the connector **2222** may include a lip that attaches to a corresponding flange, a bolted assembly, a hook and loop fastener, a screw, a latch, a threaded metal insert, and a press fit connection that attaches the shaping device **2200** to a portion of the ice shaving or snow cone machine **100**. In an implementation, the connector **2222** may be chosen from a group the group consisting of a lip that attaches to a corresponding flange, a bolted assembly, a hook and loop fastener, a screw, a latch, a threaded metal insert, and a press fit connection that attaches the shaping device **2200** to a portion of the ice shaving or snow cone machine **100**.

[0107] It will be appreciated that the shaping device **2200** and the spout assembly **115** may be manufactured from a polymeric material, including the spout **105**. The spout **105** may comprise a dome **2218** that helps shape the product, when the shaping device **2200** is disconnected from the spout assembly **115**.

[0108] In an implementation, the shaping device **2200** may comprise an attachment mechanism having a connector **2222**, wherein the connector **2222** may be a lip that attaches to a corresponding flange on the spout **105** of the spout assembly **115**.

[0109] It will be appreciated that the body **2210** of the shaping device **2200** may be any shape, including but not limited to circular shapes, oval shapes, triangular shapes, rectangular shapes, square shapes, pentagonal shapes, hexagonal shapes, octagonal shapes and other shapes that may be used to shape a shaved ice or snow cone product. Other shapes that the body may include are star, crescent, and flower shapes. In one implementation, the body **2210** may be domed shaped, such that the resulting shaved ice or snow cone product is correspondingly domed shaped. It will be appreciated that various shapes and sizes of shaping devices and bodies may be used without departing from the spirit or scope of the disclosure.

[0110] The foregoing description has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. Further, it should be noted that any or all of the aforementioned alternate implementations may be used in any combination desired to form additional hybrid implementations of the disclosure.

[0111] Further, although specific implementations of the disclosure have been described and illustrated, the disclosure is not to be limited to the specific forms or arrangements of parts so described and illustrated. The scope of the disclosure is to be defined by the claims that may be appended hereto, any future claims submitted here and in different applications, and their equivalents.

Claims

1. An ice shaving machine, comprising: a safety cover comprising an interior facing surface and an exterior facing surface; an ice block feeder; a mechanical drive mechanism; a blade assembly mechanically connected to said mechanical drive mechanism comprising a blade configured to rotate relative to a block of ice when said mechanical drive mechanism is activated; and an ice block guide that extends outward from the interior facing surface of the safety cover; wherein the ice block guide is configured to contact the block of ice while the block of ice is being shaved.
2. The ice shaving machine of claim 1, wherein the ice block guide comprises an edge; wherein the edge of the ice block guide is configured to contact the block of ice, thereby placing a compressive force on the block of ice; wherein the edge is oriented parallel to the block of ice during shaving,

such that the block of ice travels along said edge while being shaved.

3. The ice shaving machine of claim 1, wherein the ice block guide is a tensioning arm; wherein the machine further comprises a tensioning mechanism for adjusting a force that the tensioning arm exerts on the ice block, wherein the tensioning mechanism contacts a surface of the tensioning arm.
4. The ice shaving machine of claim 3, wherein the tensioning mechanism comprises a threaded member and a crank handle that is configured to drive the threaded member.
5. The ice shaving machine of claim 4, wherein the crank is hand operated.
6. The ice shaving machine of claim 4, wherein the crank is driven by an electric motor.
7. The ice shaving machine of claim 3, wherein the tensioning mechanism automatically adjusts relative to a size of the block of ice.
8. The ice shaving machine of claim 1, wherein the ice block guide is removably attached with fasteners.
9. The ice shaving machine of claim 8, wherein the fasteners are removable without tools.
10. The ice shaving machine of claim 1, wherein the ice block guide is substantially rigid and made of unitary construction and material.
11. The ice shaving machine of claim 1, wherein the ice block guide is made from a polymer material.
12. The ice shaving machine of claim 1, wherein the ice block guide comprises a blade portion that comprises an edge that is shaped to optimally contact the block of ice.
13. The ice shaving machine of claim 12, wherein the blade portion comprises a rounded profile.
14. The ice shaving machine of claim 12, wherein the blade portion comprises a sharpened edge.
15. The ice shaving machine of claim 1, wherein the blade of the blade assembly comprises a plurality of openings and a plurality of cutting surfaces adjacent to said openings for allowing ice shavings to pass therethrough, wherein said cutting surfaces and said openings are arranged on surface of said blade, and wherein said blade further comprises a plurality of attachment points; and a blade holder configured to detachably retain said blade at the plurality of attachment points, wherein said blade holder has a plurality of arms that radiate from a center hub; a control mechanism for activating and deactivating said mechanical drive mechanism.
16. The ice shaving machine of claim 15, wherein said plurality of arms comprise a pass through portion and a blade attachment portion.
17. The ice shaving machine of claim 16, wherein said center hub is configured to mechanically couple with a drive motor, wherein said center hub portion comprises a blade support for interacting with said blade.
18. The ice shaving machine of claim 1, wherein the machine further comprises a shaping device.
19. The ice shaving machine of claim 18, wherein the machine further comprises a spout assembly comprising a spout and an outer shell fitting over and around said blade.
20. The ice shaving machine of claim 19, wherein the shaping device comprises a body having an inner surface that is sized and shaped to produce a shaved ice product having a predetermined shape, such that the configuration of the inner surface of the body determines a corresponding size and shape of the shaved ice product.

21-48. (canceled)
